

# Raphaël URFIN

Centre de Sciences des Données, 45 rue d'Ulm – 75005 Paris – France

## About me

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I am a second year PhD student at École Normale Supérieure - PSL in Paris supervised by Giulio Biroli (ENS) and Marc Mézard (Bocconi University) working on the theory of diffusion models. My background is in theoretical physics.

## Education

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|------------------------------------------------------------------------------------------------|--------------------------|
| <b>PhD in Physics and Machine Learning</b>                                                     | <b>ENS-PSL, Paris</b>    |
| • <i>Subject: Generative Diffusion and Statistical Physics</i>                                 | <i>2025 - 2028(exp.)</i> |
| LPENS & Centre de Sciences des Données                                                         |                          |
| Supervisors: Giulio Biroli (ENS) and Marc Mézard(Bocconi University)                           |                          |
| <b>ENS Diploma</b>                                                                             | <b>ENS-PSL, Paris</b>    |
| • <i>Interdisciplinary Diploma validating my studies at ENS.</i>                               | <i>2021-2025</i>         |
| <b>M2 ICFP Theoretical Physics Track</b>                                                       | <b>ENS-PSL, Paris</b>    |
| • <i>First Class Honors, 17.38/20</i>                                                          | <i>2023 - 2024</i>       |
| Relevant classes: Advanced Statistical Physics, Disordered Systems, Machine Learning.          |                          |
| <b>M1 ICFP</b>                                                                                 | <b>ENS-PSL, Paris</b>    |
| • <i>First Class Honors, 17.8/20</i>                                                           | <i>2022 - 2023</i>       |
| Relevant classes: Phase Transitions, Introduction to Quantum Field Theory, General Relativity. |                          |
| <b>Bachelor in Physics and Mathematics</b>                                                     | <b>ENS-PSL, Paris</b>    |
| • <i>First Class Honors, 17.38/20 (Physics) and 16.36/20 (Mathematics)</i>                     | <i>2021 - 2022</i>       |

## Publications

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\* denotes equal contribution.

### Preprints.....

- **Raphaël Urfin\***, Tony Bonnaire\*, Giulio Biroli, Marc Mézard.  
*Double Descent and Malign Overfitting in Diffusion Models.*  
arXiv:2609.26392 (2026). <https://arxiv.org/abs/2609.26392>

### Conference Proceedings.....

- Tony Bonnaire\*, **Raphaël Urfin\***, Giulio Biroli, Marc Mézard.  
*Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.*  
NeurIPS 2025, **Best Paper Award (4 papers among +20000 submissions)**  
Advances in Neural Information Processing Systems (NeurIPS) 38, 141266-141286  
Selected for the JSTAT special issue *Machine Learning 2026: J. Stat. Mech.* (2026) 084007.

## Conferences, Workshops and Seminars

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### Invited Talks.....

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|-------------------------------------------------|-------------------------------------------------|
| <b>Generative Memory Lab, Donders Institute</b> | <b>Radboud University, Netherlands (Online)</b> |
| • <i>Invited Seminar</i>                        | <i>September 2026</i>                           |
| <b>LemanTH 2026</b>                             | <b>Lausanne, Switzerland</b>                    |
| • <i>Workshop</i>                               | <i>April 2026</i>                               |

- **Vitelli's Group Meeting** **Online**  
March 2026  
*Group meeting (non-reciprocal p-spins)*
- **Scattering Club, Data Science for Astrophysics group @ LPENS** **Online**  
February 2026  
*Invited Seminar (1 hour)*
- **Voxel51 Best of NeurIPS 2025** **Online**  
January 2026  
*Invited Seminar (30 min)*
- **Centre de Sciences des Données Internal Seminar** **Paris, France**  
December 2025  
*Invited Seminar (30 min)*
- **Machine Learning & Signal Processing @ ENS Lyon** **Lyon, France**  
November 2025  
*Invited Seminar (1 hour)*

## Contributed Talks.....

- **Journées de Physique Statistique 2026** **Paris, France**  
January 2026  
*Conference (4-min flash talk)*
- **EurIPS 2025** **Copenhagen, Denmark**  
December 2025  
*Conference (poster + 13-min oral)*
- **NeurIPS in Paris 2025** **Paris, France**  
November 2025  
*Workshop (poster + 10-min oral)*
- **StatPhys 29** **Florence, Italy**  
July 2025  
*Conference (15-min talk)*
- **Youth in High Dimensions** **Trieste, Italy**  
July 2025  
*Workshop (15-min talk)*
- **Journées de Physique Statistique 2025** **Paris, France**  
January 2025  
*Conference (4-min flash talk)*

## Summer Schools.....

- **Beg Rohu Summer School of Statistical Physics** **Beg Rohu, France**  
June 2025 & June 2026  
*Summer School*
  - **2026 Edition (Optimization and Control of Nonequilibrium Systems):** Lectures by international researchers (e.g. G. Peyré, G. Rotskoff ...). Presented a poster on Memorization/Generalization in diffusion models.
  - **2025 Edition (Learning with Machines, Physics and Mind):** Lectures on Machine Learning and Statistical Physics by international researchers (e.g., Yann LeCun, Julia Kempe, Stéphane Mallat, Marc Mézard ...). Presented a poster on Memorization/Generalization in diffusion models.
- **Complex and Glassy Systems** **Cargese, France**  
July 2024  
*Summer School*
  - Lectures on Statistical Physics and interdisciplinary applications by international researchers (e.g. Marc Mézard, Eric Vanden-Eijnden, Valentina Ros, Guy Bunin).

## Awards and Grants

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- 2025: Recipient of one of the four best paper awards at NeurIPS 2025. **(4 papers out of 20,000+ submissions)**
- 2025: CDSN, Competitive PhD scholarship for ENS students

## Reviewing

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- Machine Learning Workshops: EurIPS 2025 Workshop Principles of Generative Modeling (PriGM), ICLR 2026 Workshop DeLTa, ICLR 2026 Workshop Sci4DL.
- Machine Learning Journals: TMLR

## Skills

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- Languages: French (Native), English (Fluent), Italian (Fluent).
- Software skills: Python (Pytorch), Matlab, Latex.

## Teaching Experience

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- TA Phase Transitions (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Giulio Biroli. Fall 2026.
- TA Mathematics Tutoring for first-year students in Physics at ENS, 36 hours, taught in French with Amir-Kian Kashani-Poor. Fall 2025 and Fall 2026.
- TA Machine Learning (M2, Master ICFP, ENS-PSL), 11 hours, taught in English with Marc Lelarge and Leonardo Defilippis. Spring 2026.
- TA Stochastic Processes for Physics (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Marylou Gabri . Fall 2025.
- "Kh lles" (Preparation for the oral exams of French "Grandes  coles") in Mathematics and Physics for Classes Pr paratoires PC, Lyc e Stanislas (2022–2024).

## Relevant Experience

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- **Statistical Physics and Diffusion Models** **Bocconi University, Milano, Italy**  
*Research Internship* *February 2025-May 2025*
  - Supervisor: Marc M zard, Department of Computing Sciences.
  - Subject: The Memorization/Generalization transition in diffusion models.
  - Resulted in a publication accepted at NeurIPS 2025
- **Statistical Physics and Diffusion Models** ** cole Normale Sup rieure-PSL, France**  
*Research Internship* *April 2024-January 2025*
  - Supervisor: Giulio Biroli, Centre de Sciences des Donn es.
  - Subject: The effect of implicit regularization in diffusion models.
- **Emergent behaviors in large ecosystems** **University of Cambridge, UK**  
*Research Internship* *January-July 2023*
  - Supervisor: Camille Scalliet, Soft Matter Group, Department of Applied Mathematics and Theoretical Physics.
  - Using tools from disordered systems (Cavity Method, Random Matrix Theory...) to understand emergent collective behaviors in ecological systems.